SC-INR: Decoder-Side Sampling Consistency for Arbitrary-Scale Image Super-Resolution

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Abstract

Arbitrary-scale image super-resolution (ASISR) commonly represents an image as a local implicit function queried at continuous output coordinates. Existing Fourier implicit decoders such as LTE improve high-frequency reconstruction, but their learned cell-conditioned phase allows the output footprint to directly alter the local Fourier phase. This paper studies a more constrained decoder-side formulation: the content function is predicted from image features, while the output cell enters only through an analytic sampling response. We call this family SC-INR. For a Fourier component observed through a box footprint, the average response is a product of normalized sinc terms; SC-INR uses this response to decouple feature-conditioned content from scale-conditioned observation. Our current no-phase variant gives a stable small OOD PSNR gain over LTE across three seeds, and substantially improves same-LR cross-scale observation consistency in seed1 diagnostics. The feature-conditioned phase final candidate shows positive OOD and All-scale mean gains over LIIF and LTE in a three-seed benchmark; its comparison to the no-phase variant is reported as seed1 context.

1 Introduction

ASISR aims to recover high-resolution images at arbitrary continuous scale factors. Local implicit image functions, represented by coordinate-conditioned MLPs over image features, provide a practical route to this setting [1]. LTE improves this formulation by predicting local Fourier coefficients and frequencies from image features, enabling stronger texture modeling than a plain MLP [2].

The current work revisits how the output cell enters such Fourier decoders. In LTE, the cell is fed to a learned phase map and directly shifts the Fourier argument. This can improve fitting near training scales, but it also gives the network freedom to extrapolate cell-to-phase behavior outside the training scale range. From a sampling perspective, the cell has a more specific meaning: it is the footprint of the output pixel in the continuous domain.

We therefore study *decoder-side sampling consistency*. The central idea is to separate the feature-conditioned continuous content from the scale-conditioned observation. In SC-INR, the Fourier coefficient, frequency, and optional content phase are functions of local image features only. The output cell affects the decoded observation through an analytic sinc response derived from box averaging a Fourier component. This is not a claim of whole-network scale equivariance: the encoder, degradation model, and sampling protocol are not proven to commute with scale transformations. The contribution is instead a constrained implicit decoder that reduces arbitrary learned cell-conditioned extrapolation.

2 Method

2.1 Local Fourier Decoder

Let $E(I_{\mathrm{LR}})$ be the encoder feature map. For a query coordinate $\mathbf{x}$, the decoder samples a local feature $\mathbf{z}$ and forms a relative coordinate δ to the local feature anchor. LTE predicts coefficients and frequencies from $\mathbf{z}$, but also adds a learned phase term $h_p(\mathbf{c})$:

$$\cos\left(\pi(\boldsymbol{\omega}(\mathbf{z})^\top \delta + h_p(\mathbf{c}))\right). \tag{1}$$

This creates a cell-specific local function $f_{\mathbf{z},\mathbf{c}}$, rather than a single content function observed under different output footprints.

2.2 Analytic Observation Response

Consider a one-dimensional Fourier component

$$f(x) = A\cos(\pi\omega x + \phi). \tag{2}$$

The box-averaged observation over a cell of width c is

$$\frac{1}{c}\int_{x_0-c/2}^{x_0+c/2} A\cos(\pi\omega x + \phi)\,dx = A\cos(\pi\omega x_0 + \phi)\,\mathrm{sinc}\left(\frac{\omega c}{2}\right), \tag{3}$$

where $\mathrm{sinc}(t) = \sin(\pi t)/(\pi t)$. For a two-dimensional rectangular footprint, the separable response is

$$W(\boldsymbol{\omega}, \mathbf{c}) = \mathrm{sinc}\left(\frac{\omega_h c_h}{2}\right)\mathrm{sinc}\left(\frac{\omega_w c_w}{2}\right). \tag{4}$$

2.3 SC-INR Variants

The final-candidate SC-INR predicts content terms from features:

$$A = A(\mathbf{z}), \qquad \boldsymbol{\omega} = \boldsymbol{\omega}(\mathbf{z}), \qquad \phi = \phi(\mathbf{z}), \tag{5}$$

and uses signed bounded frequencies

$$\boldsymbol{\omega}(\mathbf{z}) = \omega_{\max}\tanh(g_\omega(\mathbf{z})). \tag{6}$$

The decoder input is

$$A(\mathbf{z}) \odot \left[\cos(\pi(\boldsymbol{\omega}(\mathbf{z})^\top\delta + \phi(\mathbf{z}))), \sin(\pi(\boldsymbol{\omega}(\mathbf{z})^\top\delta + \phi(\mathbf{z})))\right] \odot W(\boldsymbol{\omega}(\mathbf{z}), \mathbf{c}). \tag{7}$$

The feature phase $\phi(\mathbf{z})$ does not receive cell or scale. The previous no-phase variant, denoted SC-INR-NoPhi, sets $\phi = 0$. We keep checkpoint and raw-result compatibility names separate from paper display names; in older result files, the raw key "SC-INR" refers to SC-INR-NoPhi, while the current final-candidate SC-INR corresponds to raw key "SC-INR+PhiZ" in the seed1 historical file and to the cleaned raw key "SC-INR" in the seed2/seed3 files.

3 Experiments

3.1 Protocol

We use the inherited ASISR protocol with HR on-the-fly bicubic downsampling, avoiding the unreliable local pre-generated LR benchmark files. Models are trained on scales up to 4× and evaluated at in-distribution scales $\{2, 3, 4\}$ and OOD scales $\{6, 8, 12, 16, 24, 30\}$. PSNR is reported on Set5, Set14, BSD100, and Urban100. Auxiliary diagnostics are currently computed on BSD100 and Urban100 subsets for seed1.

3.2 Seed1 Final-Candidate Benchmark

Table 1: Seed1 benchmark context. The final-candidate SC-INR was strongest in seed1, but the current claim should use the three-seed summary in Table 3.

Model	ID PSNR	OOD PSNR	All PSNR
LIIF	30.9992	22.7111	25.4738
LTE	31.0450	22.6917	25.4761
SC-INR-NoPhi	31.0515	22.7471	25.5152
SC-INR-NoPhi-Signed	31.0567	22.7418	25.5134
SC-INR	31.1187	22.7766	25.5573
LIIF-EQ	31.1205	22.7469	25.5381

Table 1 shows that the final-candidate SC-INR improves seed1 OOD PSNR by +0.0849 dB over LTE and by +0.0295 dB over SC-INR-NoPhi. This is the intended place to show the SC-INR vs SC-INR-NoPhi structural increment.

3.3 Core Multi-Seed Result

Table 2: Core three-seed benchmark for LIIF, LTE, and SC-INR-NoPhi. The stable multi-seed evidence currently belongs to the no-phase variant.

Model	Split	Mean PSNR	Std.	Delta vs LTE
LIIF	ID	31.0301	0.0274	-0.0485
LIIF	OOD	22.7251	0.0124	+0.0177
LTE	ID	31.0786	0.0296	+0.0000
LTE	OOD	22.7074	0.0137	+0.0000
SC-INR-NoPhi	ID	31.0746	0.0203	-0.0040
SC-INR-NoPhi	OOD	22.7581	0.0105	+0.0507

Table 2 indicates that SC-INR-NoPhi gives a stable small OOD PSNR gain over LTE, while its ID average is essentially tied with LTE. This is consistent with the intended role of decoder-side sampling consistency: improving scale extrapolation behavior rather than merely increasing training-scale fitting.

3.4 Final-Candidate Multi-Seed Result

Table 3: Final-candidate SC-INR three-seed benchmark. Deltas are paired seed deltas against LIIF and LTE under the same benchmark protocol.

Split	Mean PSNR	Std.	Delta vs LIIF	Delta vs LTE
ID	31.0738	0.0582	+0.0437	-0.0048
OOD	22.7578	0.0316	+0.0328	+0.0504
All	25.5298	0.0403	+0.0364	+0.0320

Table 3 shows that the final-candidate SC-INR retains the OOD and All-scale PSNR advantage over LTE and is positive over LIIF across ID, OOD, and All splits. We keep SC-INR vs SC-INR-NoPhi as seed1 context rather than foregrounding seed2/seed3 differences between those two internal variants.

3.5 Cross-Scale Observation Consistency

Table 4: Seed1 same-LR cross-scale observation consistency. Larger consistency PSNR indicates closer agreement when outputs at larger scales are compared to the 4× observation under the same LR input.

Model	Dataset	Consistency PSNR	Delta vs LTE
LTE	BSD100	44.1914	+0.0000
SC-INR-NoPhi	BSD100	52.8053	+8.6140
LTE	Urban100	39.2983	+0.0000
SC-INR-NoPhi	Urban100	47.0694	+7.7712

The large consistency gains in Table 4 support the mechanism that removing learned cell-conditioned phase and using an analytic observation response improves same-LR cross-scale agreement. However, diagnostic LTE variants with weakened cell response can also score highly on consistency; these numbers must be interpreted jointly with reconstruction quality.

3.6 Selected Qualitative Examples

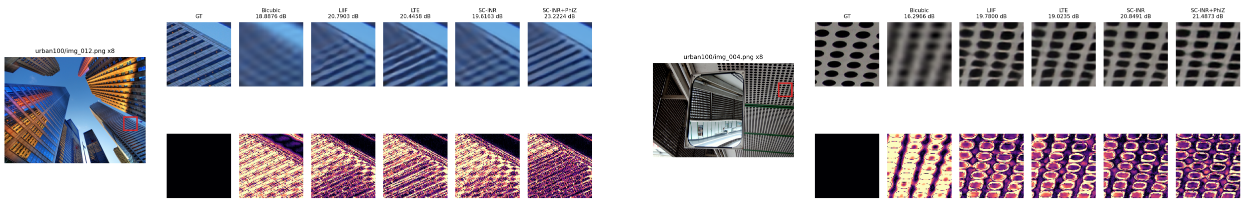

Figure 1: User-confirmed selected Urban100 8× examples where the final-candidate SC-INR shows clearer local structure than LTE. These are selected examples, not evidence of average visual quality.

4 Limitations and Next Steps

The current evidence is not yet a complete paper claim. First, final-candidate SC-INR now has three-seed benchmark evidence against LIIF and LTE, but auxiliary metrics for the final candidate remain seed1-only. Second, the w/o-sinc ablation is still single-seed and should not be used as a unique causal proof of the analytic response. Third, same-LR consistency can reward cell-insensitive behavior, so footprint-correctness diagnostics remain necessary. Finally, PSNR does not fully measure perceptual texture quality; qualitative crops and texture-sensitive metrics must remain part of the evidence package.

5 Conclusion

SC-INR reframes output scale in Fourier implicit ASISR decoders as an observation footprint rather than a learned phase shortcut. The current results support decoder-side sampling consistency as a useful principle: SC-INR-NoPhi shows stable small OOD gains and substantially better cross-scale observation consistency, while the final-candidate feature-phase SC-INR preserves the OOD and All-scale PSNR gain over LTE and improves over LIIF on average. The next critical step is to close the remaining evidence gap with final-candidate auxiliary metrics, footprint-correctness diagnostics, and carefully audited qualitative evidence.

References

[1] Yinbo Chen, Sifei Liu, and Xiaolong Wang. Learning continuous image representation with local implicit image function. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8628–8638, 2021.

[2] Jaewon Lee and Kyong Hwan Jin. Local texture estimator for implicit representation function. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1929–1938, 2022.